

Python Data Science Model Evidence

Same problem (overcrowding prediction) and same feature set as in Section 5, trained independently in scikit-learn/XGBoost pipeline , Spark results are never used as input for Python.

Preprocessing and Feature Engineering

Shares the trip_features table generated by the Feature Engineering step; one-hot and label encodings of categorical variables happen inside the Python pipeline and not using Spark pipeline.

Algorithms, Hyperparameters, and Results

Model	Hyperparameters	Val. AUC	Test AUC	Precision	Recall	F1	Accuracy
LogisticRegression	max_iter=200	0.9579	0.9555	0.8044	0.7937	0.7990	0.9073
RandomForest	max_depth=14, n_estimators=120	0.9543	0.9523	0.9626	0.4185	0.5834	0.8612
XGBoost	colsample_bytree=0.9, lr=0.1, max_depth=7, n_estimators=250, subsample=0.9	0.9835	0.9830	0.9028	0.8077	0.8526	0.9351

Best model selected on validation AUC: XGBoost (0.984) , confirmed on test at 0.983 AUC / 0.853 F1.

Accuracy, Precision, Recall, F1-score

Reported per model in the table above; confusion matrix for the selected model:

	Predicted: Not Overcrowded	Predicted: Overcrowded
Actually Not Overcrowded	114,309	3,088
Actually Overcrowded	6,829	28,683

Forecast Metrics

Comparison between demand forecasting (HistGradientBoostingRegressor) and the same weekday naive benchmark, both evaluated using the same chronological test data:

Model	MAE	RMSE	MAPE	R ²
HistGBR (ours)	156.85	237.98	28.7%	0.8489
Baseline (last week)	178.25	276.33	29.28%	0.7962

Occupancy forecasting comparison:

Model	MAE	RMSE	MAPE
Occupancy model	0.113	0.152	24.69%
Baseline (last week)	0.139	0.191	29.13%

Saved Model and Sample Predictions

Same versioned storage system as in Section 5.4 (src/utiq/registry.py) within the Python pipeline model namespace.